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SOLID TARGET SUBSTANCE REPLENISHMENT DEVICE, EXTREME ULTRAVIOLET LIGHT GENERATION APPARATUS, AND ELECTRONIC DEVICE MANUFACTURING METHOD

Abstract

A solid target substance replenishment device includes a solid target container, a first path through which a solid target substance passes, a delivery device, and a second path through which the solid target substance delivered by the delivery device is replenished to a molten target container. The delivery device includes a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface; a delivery rod delivering the solid target substance; and a stopper forming a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of Japanese Patent Application No. 2024-025770, filed on Feb. 22, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a solid target substance replenishment device, an extreme ultraviolet light generation apparatus, and an electronic device manufacturing method.

2. Related Art

[0003] Recently, miniaturization of a transfer pattern in optical lithography of a semiconductor process has been rapidly proceeding along with miniaturization of the semiconductor process. In the next generation, microfabrication at 10 nm or less will be required. Therefore, it is expected to develop an exposure apparatus that combines an extreme ultraviolet (EUV) light generation apparatus that generates EUV light having a wavelength of about 13 nm and reduced projection reflection optical system.

[0004] As the EUV light generation apparatus, a laser produced plasma (LPP) type apparatus using plasma generated by irradiating a target substance with laser light has been developed.

LIST OF DOCUMENTS

Patent Documents

[0005] Patent Document 1: US Patent Application Publication No. 2023/0008733 [0006] Patent Document 2: Japanese Patent Application Publication No. H07-051419 [0007] Patent Document 3: Japanese Patent Application Publication No. 2003-164553 [0008] Patent Document 4: Japanese Patent Application Publication No. H10-146452 [0009] Patent Document 5: Japanese Patent Application Publication No. H08-071208 [0010] Patent Document 6: Japanese Patent Application Publication No. 2007-020646

SUMMARY

[0011] A solid target substance replenishment device according to an aspect of the present disclosure includes a solid target container configured to contain a solid target substance, a first path through which the solid target substance supplied from the solid target container passes, a delivery device, and a second path through which the solid target substance delivered by the delivery device is replenished to a molten target container of an extreme ultraviolet light generation apparatus. Here, the delivery device includes a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward.

[0012] An electronic device manufacturing method according to an aspect of the present disclosure

includes generating extreme ultraviolet light using an extreme ultraviolet light generation apparatus, outputting the extreme ultraviolet light to an exposure apparatus, and exposing a photosensitive substrate to the extreme ultraviolet light in the exposure apparatus to manufacture an electronic device. Here, the extreme ultraviolet light generation apparatus includes a solid target substance replenishment device; a molten target container configured to melt the solid target substance replenished by the solid target substance replenishment device to produce a molten target substance; a nozzle configured to output the molten target substance produced in the molten target container; a laser device configured to irradiate, with pulse laser light, the molten target substance having reached a predetermined region after being output from the nozzle; and an EUV light concentrating mirror configured to concentrate the extreme ultraviolet light emitted from plasma generated in the predetermined region. The solid target substance replenishment device includes a solid target container configured to contain a solid target substance, a first path through which the solid target substance supplied from the solid target container passes, a delivery device, and a second path through which the solid target substance delivered by the delivery device is replenished to the molten target container of the extreme ultraviolet light generation apparatus. The delivery device includes a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward.

[0013] An electronic device manufacturing method according to an aspect of the present disclosure includes inspecting a defect of a mask by irradiating the mask with extreme ultraviolet light generated by an extreme ultraviolet light generation apparatus, selecting a mask using a result of the inspection, and exposing and transferring a pattern formed on the selected mask onto a photosensitive substrate. Here, the extreme ultraviolet light generation apparatus includes a solid target substance replenishment device, a molten target container configured to melt the solid target substance replenished by the solid target substance replenishment device to produce a molten target substance, a nozzle configured to output the molten target substance produced in the molten target container, a laser device configured to irradiate, with pulse laser light, the molten target substance having reached a predetermined region after being output from the nozzle, and an EUV light concentrating mirror configured to concentrate the extreme ultraviolet light emitted from plasma generated in the predetermined region. The solid target substance replenishment device includes a solid target container configured to contain a solid target substance, a first path through which the solid target substance supplied from the solid target container passes; a delivery device, and a second path through which the solid target substance delivered by the delivery device is replenished to the molten target container of the extreme ultraviolet light generation apparatus. The delivery device includes a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] Embodiments of the present disclosure will be described below merely as examples with reference to the accompanying drawings.

[0015] FIG. 1 shows the configuration of an LPP EUV light generation system.

[0016] FIG. 2 shows the configuration of a droplet target generation device in the EUV light generation system according to a comparative example.

[0017] FIG. 3 shows the configuration and operation of delivery device according to the comparative example.

[0018] FIG. 4 shows the configuration and operation of the delivery device according to the comparative example.

[0019] FIG. 5 shows the configuration and operation of the delivery device according to the comparative example.

[0020] FIG. 6 shows the configuration and operation of the delivery device according to the comparative example.

[0021] FIG. 7 shows a state in the middle of forward movement of a delivery rod in the comparative example.

[0022] FIG. 8 shows a state in the middle of the forward movement of the delivery rod in the comparative example.

[0023] FIG. 9 schematically shows the configuration of the delivery device according to a first embodiment.

[0024] FIG. 10 shows the configuration of the delivery device according to the first embodiment.

[0025] FIG. 11 shows the operation of the delivery device according to the first embodiment.

[0026] FIG. 12 shows the operation of the delivery device according to the first embodiment.

[0027] FIG. 13 shows the operation of the delivery device according to the first embodiment.

[0028] FIG. 14 shows the operation of the delivery device according to the first embodiment.

[0029] FIG. 15 shows the configuration of the delivery device according to a second embodiment.

[0030] FIG. 16 is a sectional view of a tapered portion of a stopper according to a first modification of the second embodiment.

[0031] FIG. 17 is a sectional view of the tapered portion of the stopper according to a second modification of the second embodiment.

[0032] FIG. 18 shows the configuration of a droplet target generation device of a third embodiment.

[0033] FIG. 19 shows the configuration of an exposure apparatus connected to the EUV light generation system.

[0034] FIG. 20 shows the configuration of an inspection apparatus connected to the EUV light generation system.

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which first surface **821** includes groove [0055] 5.1 Configuration and operation [0056] 5.2 Effect [0057] 6. Solid target substance replenishment device **260c** including first and second detectors **D1**, **D2** [0058] 6.1 Configuration [0059] 6.2 Operation [0060] 6.3 Effect [0061] 7. Others [0062] 7.1 EUV light utilization apparatus **6** [0063] 7.2 Supplement [0064] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. The embodiments described below show some examples of the present disclosure and do not limit the contents of the present disclosure. Also, all configurations and operation described in the embodiments are not necessarily essential as configurations and operation of the present disclosure. Here, the same components are denoted by the same reference numeral, and duplicate description thereof is omitted.

1. Overall Description of EUV Light Generation System **11**

1.1 Configuration

[0065] FIG. **1** shows the configuration of an LPP EUV light generation system **11**. An EUV light generation apparatus **1** is used together with a laser device **3**. In the present disclosure, a system including the EUV light generation apparatus **1** and the laser device **3** is referred to as the EUV light generation system **11**. The EUV light generation apparatus **1** includes a chamber **2** and a droplet target generation device **26**. The chamber **2** is a sealable container. The droplet target generation device **26** supplies a target **27** containing a target substance into the chamber **2**. The material of the target substance may include tin, terbium, gadolinium, lithium, xenon, or a combination of any two or more thereof.

[0066] A through hole is formed in a wall of the chamber **2**. The through hole is blocked by a window **21** and pulse laser light **32** output from the laser device **3** is transmitted through the window **21**. An EUV light concentrating mirror **23** having a spheroidal reflection surface is arranged in the chamber **2**. The EUV light concentrating mirror **23** has first and second focal points. A multilayer reflection film in which molybdenum and silicon are alternately stacked is formed on a surface of the EUV light concentrating mirror **23**. The EUV light concentrating mirror **23** is arranged such that the first focal point is located in a plasma generation region **25** and the second focal point is located at an intermediate focal point **292**. A through hole **24** is formed at the center of the EUV light concentrating mirror **23**, and pulse laser light **33** passes through the through hole **24**.

[0067] The EUV light generation apparatus **1** includes an EUV light generation processor **5**, a target sensor **4**, and the like. The EUV light generation processor **5** is a processing device including a memory **501** in which a control program is stored and a central processing unit (CPU) **502** which executes the control program. The EUV light generation processor **5** is specifically configured or programmed to perform various processes included in the present disclosure. The target sensor **4** detects at least one of the presence, trajectory, position, and velocity of the target **27**. The target sensor **4** may have an imaging function.

[0068] Further, the EUV light generation apparatus **1** includes a connection portion **29** providing communication between the internal space of the chamber **2** and the internal space of an EUV light utilization apparatus **6**. The EUV light utilization apparatus **6** may be an exposure apparatus **6a** shown in FIG. **19** or an inspection apparatus **6b** shown in FIG. **20**. A wall **291** in which an aperture is formed is arranged in the connection portion **29**. The wall **291** is arranged such that the aperture is located at the second focal point of the EUV light concentrating mirror **23**.

[0069] Further, the EUV light generation apparatus **1** includes a laser light transmission device **34**, a laser light concentrating mirror **22**, a target collection unit **28** for collecting the target **27**, and the like. The laser light transmission device **34** includes an optical element for defining a transmission state of the pulse laser light **32**, and an actuator for adjusting the position, posture, and the like of the optical element.

1.2 Operation

[0070] Operation of the EUV light generation system **11** will be described with reference to FIG. **1**.

Pulse laser light **31** output from the laser device **3** enters, via the laser light transmission device **34**, the chamber **2** through the window **21** as the pulse laser light **32**. The pulse laser light **32** travels along a laser light path in the chamber **2**, is reflected by the laser light concentrating mirror **22**, and is radiated to the target **27** as the pulse laser light **33**.

[0071] The droplet target generation device **26** outputs the target **27** toward the plasma generation region **25** in the chamber **2**. The target **27** is irradiated with the pulse laser light **33**. The target **27** irradiated with the pulse laser light **33** is turned into plasma, and radiation light **251** is radiated from the plasma. The EUV light contained in the radiation light **251** is reflected by the EUV light concentrating mirror **23** with higher reflectance than light in other wavelength ranges. Reflection light **252** including the EUV light reflected by the EUV light concentrating mirror **23** is concentrated at the intermediate focal point **292** and output to the EUV light utilization apparatus **6**. Here, one target **27** may be irradiated with a plurality of pulses included in the pulse laser light **33**. [0072] The EUV light generation processor **5** controls the entire EUV light generation system **11**. The EUV light generation processor **5** processes a detection result of the target sensor **4**. Based on the detection result of the target sensor **4**, the EUV light generation processor **5** controls the timing at which the target **27** is output, the output direction of the target **27**, and the like. Further, the EUV light generation processor **5** controls an oscillation timing of the laser device **3**, the travel direction of the pulse laser light **32**, the concentration position of the pulse laser light **33**, and the like. Such various kinds of control described above are merely exemplary, and other control may be added as necessary.

2. Comparative Example

2.1 Configuration of Droplet Target Generation Device **26**

[0073] FIG. **2** shows the configuration of the droplet target generation device **26** in the EUV light generation system according to the comparative example. The comparative example of the present disclosure is an example recognized by the applicant as known only by the applicant, and is not a publicly known example admitted by the applicant. The droplet target generation device **26** includes a solid target substance replenishment device **260**, a molten target container **C3**, and a nozzle **52**.

2.1.1 Solid Target Substance Replenishment Device **260**

[0074] The solid target substance replenishment device **260** includes a solid target container **C1**, a feed device **7**, a delivery device **8**, a funnel **9**, a load lock chamber **C2**, a target supply processor **55**, supply pipes **40** to **45**, a gas cylinder **G1**, and a pressure regulator **56**. The path of the solid target substance **270** configured by the feed device **7** and the supply pipe **40** corresponds to a first path through which the solid target substance **270** supplied from the solid target container **C1** passes. The path of the solid target substance **270** from the supply pipe **41** to the supply pipe **45** corresponds to a second path for replenishing the solid target substance **270** delivered by the delivery device **8** to the molten target container **C3**.

[0075] The target supply processor **55** is a processing device including a memory **551** in which a control program is stored and a CPU **552** which executes the control program. The target supply processor **55** corresponds to the processor in the present disclosure. The target supply processor **55** is specifically configured or programmed to perform various processes included in the present disclosure.

[0076] The solid target container **C1** is a container to contain the solid target substance **270** such as tin. The solid target substance **270** may be, for example, spherical particles of substantially the same size. The diameter of the solid target substance **270** is, for example, 2 mm or more and 5 mm or less. The temperature in the solid target container **C1** is lower than the melting point of the target substance. The pressure in the solid target container **C1** is about the same as the atmospheric pressure.

[0077] The feed device **7** is connected to the bottom of the solid target container **C1**, and connected to the delivery device **8** via the supply pipe **40**. Details of the delivery device **8** will be described

later with reference to FIGS. 3 to 6. The supply pipe **41** is arranged obliquely with respect to the gravity direction, and the delivery device **8** is connected to the lower end of the supply pipe **41**. The funnel **9** is arranged below a discharge port **410** near the upper end of the supply pipe **41** as being spaced apart from the discharge port **410**.

[0078] The funnel **9** has a conical shape whose diameter decreases toward the gravity direction, and the lower end thereof is connected to the load lock chamber C2 via the supply pipes **42, 43**. A valve **V1** is connected between the supply pipes **42, 43**.

[0079] The load lock chamber C2 is a container to contain the solid target substance **270** supplied from the solid target container C1. The temperature in the load lock chamber C2 is lower than the melting point of the target substance. The load lock chamber C2 is connected to the molten target container C3 via the supply pipes **44, 45**. A valve **V2** is connected between the supply pipes **44, 45**.

2.1.2 Molten Target Container C3

[0080] The molten target container C3 contains the target substance supplied from the load lock chamber C2. The molten target container C3 is connected to a gas cylinder G1 via a pressurized gas pipe LO. The gas cylinder G1 contains a high-pressure rare gas such as an argon gas or a helium gas as a pressurized gas. A pressure regulator **56** and a pressure gauge P is arranged at the pressurized gas pipe LO. The target supply processor **55** controls the pressure regulator **56** based on an output of the pressure gauge P, so that the pressure in the molten target container C3 is adjusted to a predetermined pressure higher than the atmospheric pressure.

[0081] A heater **51** and a level sensor **54** are arranged at the molten target container C3. The heater **51** is connected to a power source (not shown) and heats the inside of the molten target container C3 to a predetermined temperature higher than the melting point of the target substance. The temperature in the molten target container C3 is controlled by controlling the power source based on the output of a temperature sensor (not shown) arranged at the molten target container C3. As a result, the solid target substance **270** is melted in the molten target container C3 to generate a molten target substance. The level sensor **54** detects a liquid level position of the molten target substance in the molten target container C3.

2.1.3 Nozzle 52

[0082] The nozzle **52** is arranged at a lower end of the molten target container C3 in the gravity direction. The tip of the nozzle **52** is opened to the inside of the chamber 2 (see FIG. 1). A piezoelectric element **53** is arranged at the nozzle **52**.

2.2 Operation of Droplet Target Generation Device 26

[0083] When the level sensor **54** detects that the liquid level position of the molten target substance in the molten target container C3 becomes below a threshold, the solid target substance **270** is replenished from the solid target container C1 as described below.

[0084] The target supply processor **55** opens the valve **V1** with the valve **V2** closed. Since the valve **V2** is closed, the inside of the molten target container C3 is maintained at a high pressure. By opening the valve **V1**, the load lock chamber C2 is ready to receive the solid target substance **270**.

[0085] The target supply processor **55** calculates the supply amount of the solid target substance **270** by the feed device 7 from the shortage amount of the molten target substance in the molten target container C3, and outputs a feed signal to the feed device 7. The feed device 7 feeds the solid target substance **270** one by one to the supply pipe **40** according to a feed signal. The target supply processor **55** does not output a subsequent feed signal until the replenishment of the solid target substance **270** to the molten target container C3 based on the latest feed signal is completed.

[0086] The delivery device **8** receives the solid target substance **270** from the supply pipe **40**, and delivers the solid target substance **270** to the supply pipe **41** one by one at regular time intervals according to a control signal from the target supply processor **55**.

[0087] The solid target substance **270** delivered from the delivery device **8** to the supply pipe **41** one by one is pressed by the solid target substance **270** delivered subsequently. As a result, the plurality of solid target substances **270** are successively moved in the supply pipe **41** against

gravity, and are discharged from the discharge port **410** in order from the solid target substance **270** at the top.

[0088] The solid target substance **270** discharged from the discharge port **410** is received by the funnel **9** and flows into the supply pipe **42**. The solid target substance **270** moves to the load lock chamber C2 through the open valve V1 and the supply pipe **43**. When a desired amount of the solid target substances **270** is moved to the load lock chamber C2, the target supply processor **55** stops operation of the feed device **7** and the delivery device **8** and closes the valve V1.

[0089] Next, the target supply processor **55** opens the valve V2 to replenish the solid target substance **270** contained in the load lock chamber C2 to the molten target container C3. The solid target substance **270** moves from the load lock chamber C2 to the molten target container C3. The solid target substance **270** supplied to the molten target container C3 melts and mixes with the target substance already contained and melted in the molten target container C3. The heater **51** suppresses a decrease in the internal temperature of the molten target container C3.

[0090] When the valve V2 is opened, a portion of the gas in the molten target container C3 moves to the load lock chamber C2, and the pressure inside the molten target container C3 temporarily decreases. Since the valve V1 is closed before opening the valve V2, the high-pressure gas in the molten target container C3 is prevented from flowing from the valve V1 towards the funnel **9**.

Further, the pressurized gas in the gas cylinder G1 is supplied to the molten target container C3 via the pressure regulator **56**, whereby the pressure in the molten target container C3 is recovered.

[0091] The molten target substance in the molten target container C3 is output from the opening at the tip of the nozzle **52** owing to the pressure difference between the pressurized gas supplied from the pressure regulator **56** and the pressure in the chamber 2. When vibration is applied to the nozzle **52** by the piezoelectric element **53**, the jet-like molten target substance output from the nozzle **52** is separated into droplets to form the target **27**.

[0092] According to the comparative example, the solid target substance **270** contained in the solid target container C1 having a substantially atmospheric pressure can be supplied into the molten target container C3 having a high pressure. Even when the target substance in the molten target container C3 is consumed, the target substance can be replenished without replacing the molten target container C3, so that the downtime of the EUV light generation apparatus **1** can be reduced.

2.3 Configuration of Delivery Device **8**

[0093] FIGS. **3** to **6** show the configuration and operation of the delivery device **8** according to the comparative example. In FIGS. **3** to **6**, the individual solid target substances **270** are denoted by the reference signs α to ω , and the reference numerals α to ω may be used to distinguish them. The delivery device **8** includes a tube **80**, a delivery rod **81**, a stopper **82**, and a base portion **83**.

[0094] The tube **80** has a cylindrical shape having a straight center axis, and includes a receiving port **801** for receiving the solid target substance **270** having passed through the feed device **7** and the supply pipe **40**, and an entrance **802** facing the receiving port **801**. The delivery rod **81** is located inside the tube **80** and configured to deliver the solid target substance **270** in the tube **80** by being caused to alternately move forward and backward in the length direction of the tube **80** by a drive unit **84**.

[0095] The direction of the forward movement of the delivery rod **81** is defined as an X direction, the direction in which the solid target substance **270** passes through the receiving port **801** is defined as a Z direction, and one of the directions perpendicular to the X direction and the Z direction is defined as a Y direction. FIGS. **3** to **6** are views of the delivery device **8** viewed in the -Y direction. The tube **80** is supported by the base portion **83** at an angle A with respect to the horizontal direction, and connected to the supply pipe **41** having a larger inclination. The drive unit **84** is supported by the base portion **83**.

[0096] The stopper **82** includes a tapered portion **820** and a rod portion **824**. The tapered portion **820** is supported by the rod portion **824**, and the rod portion **824** is supported by the base portion **83** via a rotation shaft **825**. The rotation shaft **825** is parallel to the Y direction, and the stopper **82**

is movable while pivoting in a plane parallel to an XZ plane.

[0097] The tapered portion **820** passes through the entrance **802** of the tube **80**. The tapered portion **820** includes a first surface **821** and a second surface **822**. The first surface **821** is inclined with respect to the X direction, and the second surface **822** intersects the X direction at an angle closer to perpendicular thereto than the angle between the X direction and the first surface **821**.

[0098] The stopper **82** is provided with a counterclockwise rotational force in FIGS. **3** to **6** by the restoring force of the spring **826**. When the delivery rod **81** moves forward in the X direction, the stopper **82** is pressed by the delivery rod **81** and pivots clockwise against the restoring force of the spring **826**. When the delivery rod **81** is moved backward in the $-X$ direction, the stopper **82** pivots counterclockwise by the restoring force of the spring **826**, but when the stopper **82** abuts a stop pin **827**, the counterclockwise pivot of the stopper **82** is restricted. At this time, the stopper **82** restricts the solid target substance δ in the tube **80** from returning in the $-X$ direction. A space for receiving the solid target substance γ supplied from the receiving port **801** is secured between the receiving port **801** and the entrance **802**.

2.4 Operation of Delivery Device **8**

[0099] Referring to FIGS. **3** to **6**, description is provided on the operation of the delivery device **8** to deliver the solid target substance γ waiting in the supply pipe **40** to the supply pipe **41**. As a result of the delivery device **8** delivering the plurality of solid target substances **270** one by one, as shown in FIG. **3**, it is assumed that the solid target substances δ to ω are already filled in the tube **80** and the supply pipe **41**. When the delivery rod **81** moves backward to its most retracted position in the $-X$ direction, the solid target substance γ moves into the space between the receiving port **801** and the entrance **802**.

[0100] Thereafter, the delivery rod **81** moves in the X direction in the tube **80** and presses the first surface **821** of the stopper **82** via the solid target substance γ between the receiving port **801** and the entrance **802**. As shown in FIG. **4**, the stopper **82** pressed by the delivery rod **81** pivots clockwise in FIG. **4** against the restoring force of the spring **826**, and most of the tapered portion **820** moves to the outside of the tube **80**. Thus, a passage for moving the solid target substance γ in the X direction is formed in the tube **80**.

[0101] The solid target substance γ is pressed by the delivery rod **81** moving in the X direction while being pressed against the wall surface on the $-Z$ direction side in the tube **80** by receiving the reaction force from the inclined first surface **821** of the stopper **82**, and moves in the tube **80** as indicated by an arrow B in FIG. **4**. The solid target substance γ presses the solid target substances δ to ω . In this way, the delivery rod **81** delivers the solid target substances γ to ω against gravity, causing the solid target substance ω to be discharged from the discharge port **410**.

[0102] As the solid target substance γ moves away from the first surface **821** by passing over the ridge line between the first surface **821** and the second surface **822** of the tapered portion **820**, the stopper **82** pivots counterclockwise in FIG. **4** to the distal end position of the delivery rod **81** by the restoring force of the spring **826**. That is, the passage of the solid target substance **270** in the tube **80** is maximized when the solid target substance γ passes over the ridge line between the first surface **821** and the second surface **822** of the tapered portion **820**, and thereafter the tapered portion **820** moves toward the inside of the tube **80**.

[0103] As shown in FIG. **5**, when the delivery rod **81** starts moving in the $-X$ direction, the stopper **82** pivots counterclockwise in FIG. **5** while being pressed against the delivery rod **81** by the restoring force of the spring **826**, and the tapered portion **820** further moves toward the inside of the tube **80**. At this time, since the solid target substance γ cannot pass over the ridge line between the first surface **821** and the second surface **822**, it cannot return to the space between the receiving port **801** and the entrance **802**, and is in contact with the second surface **822**. The counterclockwise pivot of the stopper **82** is stopped by the stop pin **827**.

[0104] As shown in FIG. **6**, when the delivery rod **81** is moved to the position most retracted in the $-X$ direction, the solid target substance β waiting in the supply pipe **40** moves to the space between

the receiving port **801** and the entrance **802**.

[0105] Thereafter, by repeating the operation described with reference to FIGS. **3** to **6**, the solid target substance **270** is delivered one by one to the supply pipe **41**.

2.5 Problem of Comparative Example

[0106] FIGS. **7** and **8** show a state in the middle of the forward movement of the delivery rod **81** in the comparative example. As described with reference to FIG. **4**, the solid target substance γ pressed by the delivery rod **81** is pressed against the wall surface in the tube **80** by receiving the reaction force from the first surface **821**. In some cases, the solid target substance γ is pressed against the edge of the receiving port **801**, so that the solid target substance γ is surrounded by the edge of the receiving port **801**, the delivery rod **81**, and the first surface **821** to be immovable. The phenomenon that the solid target substance **270** becomes immovable in this manner is hereinafter referred to as “biting.”

[0107] The size of the solid target substance **270** is not completely uniform, and the shape thereof is not perfect sphere, and there are variations. As shown in FIG. **7**, a large-diameter solid target substance γ may cause biting. As shown in FIG. **8**, a non-spherical solid target substance γ may cause biting.

[0108] In each of FIGS. **7** and **8**, if the delivery rod **81** is moved in the X direction with a sufficient force, a portion of the surface of the solid target substance γ is scraped off by the edge of the receiving port **801**, and the biting may be eliminated. However, since the solid target substance γ is deformed when a portion of the surface of the solid target substance γ is scraped off, the deformed solid target substance γ may again cause biting or clogging somewhere in the second path. Further, scraping debris generated when a portion of the surface of the solid target substance γ is scraped may adhere to the rotation shaft **825** of the stopper **82** to increase friction, which may inhibit the operation of the stopper **82**. The scraping debris may adhere to the inner wall surface of the tube **80** or the supply pipe **41**, and clogging of the solid target substance **270** may occur.

[0109] Embodiments described below relate to stably replenishing the solid target substance **270** even when there are variations in the diameter and shape of the solid target substance **270**.

3. Delivery Device **8a** with Stopper **82** Inclined

3.1 Configuration of Delivery Device **8a**

[0110] FIGS. **9** and **10** schematically show the configuration of the delivery device **8a** according to a first embodiment. FIG. **9** is a view of the delivery device **8a** viewed in the $-Y$ direction, and FIG. **10** is a view of the delivery device **8a** viewed in the Z direction. The tube **80** includes a facing surface **803** facing the receiving port **801**. The facing surface **803** is a surface without the entrance **802** through which the stopper **82** enters and exits, and may be a cylindrical surface. The entrance **802** is not located at a position facing the receiving port **801**, but is located on the Y-direction side perpendicular to both the direction in which the receiving port **801** and the facing surface **803** face each other and the length direction of the tube **80**. The entrance **802** has a rectangular shape that is longer in the X direction than in the Z direction.

[0111] The stopper **82** is located on the Y direction side of the tube **80**. The rotation shaft **825** is, for example, parallel to the Z direction, and the stopper **82** is pivotable in a plane intersecting the Z direction and pivotable in a plane away from the receiving port **801**. More preferably, the stopper **82** is pivotable in a plane perpendicular to the Z direction.

[0112] The tapered portion **820** passes through the entrance **802**. The stopper **82** is provided with a counterclockwise rotational force in FIG. **10** by the restoring force of the spring **826**. The counterclockwise pivot of the stopper **82** is restricted by the stop pin **827**.

3.2 Operation of Delivery Device **8a**

[0113] FIGS. **11** to **14** schematically show the operation of the delivery device **8a** according to the first embodiment. FIGS. **11** to **14** are views of the delivery device **8a** viewed in the Z direction, and show substantially the same operation as that of FIGS. **3** to **6**, except that the stopper **82** is pivoted in a plane perpendicular to the Z direction.

[0114] When the delivery rod **81** moves forward in the X direction, the first surface **821** of the tapered portion **820** is pressed by the delivery rod **81** via the solid target substance γ , and the stopper **82** is moved toward the outside of the tube **80** by pivoting clockwise in FIGS. **11** to **14**. Thus, the passage for moving the solid target substance γ between the receiving port **801** and the facing surface **803** in the tube **80** is formed.

[0115] Since the stopper **82** pivots in a plane perpendicular to the Z direction, the solid target substance γ moves in the tube **80** as indicated by an arrow E in FIG. **12** while being pressed against the wall surface on the $-Y$ direction side in the tube **80**. Since pressing against the wall surface on the $-Z$ direction side is prevented, biting is suppressed even with the large-diameter solid target substance γ .

[0116] When the delivery rod **81** is moved backward in the $-X$ direction, the stopper **82** pivots counterclockwise by the restoring force of the spring **826** and is moved toward the inside of the tube **80**, but when the stopper **82** abuts the stop pin **827**, the counterclockwise pivot of the stopper **82** is restricted. At this time, the second surface **822** restricts the solid target substance γ in the tube **80** from returning in the $-X$ direction. Further, by further moving the delivery rod **81** in the $-X$ direction, a space for receiving the solid target substance γ supplied from the receiving port **801** is secured between the receiving port **801** and the facing surface **803** and between the delivery rod **81** and the stopper **82**.

3.3 Effect

[0117] (1) According to the first embodiment, the solid target substance replenishment device **260** includes the solid target container **C1**, the feed device **7**, the supply pipe **40**, the delivery device **8a**, and the path from the supply pipe **41** to the supply pipe **45**. The solid target container **C1** contains the solid target substance **270**. The solid target substance **270** supplied from the solid target container **C1** passes through the feed device **7** and the supply pipe **40**. The delivery device **8a** includes the tube **80**, the delivery rod **81**, and the stopper **82**. The tube **80** includes the receiving port **801** for receiving the solid target substance **270** having passed through the feed device **7** and the supply pipe **40**, and the facing surface **803** facing the receiving port **801**. The delivery rod **81** delivers the solid target substance **270** in the tube **80** by being caused to alternately move forward and backward in parallel to the X direction being the length direction of the tube **80**. When the delivery rod **81** moves forward in the X direction, the stopper **82** moves toward the outside of the tube **80**, thereby forming a passage through which the solid target substance **270** between the receiving port **801** and the facing surface **803** is moved in the tube **80**. When the delivery rod **81** moves backward in the $-X$ direction, the stopper **82** moves toward the inside of the tube **80**, thereby restricting the solid target substance **270** in the tube **80** from returning between the receiving port **801** and the facing surface **803**. The path from the supply pipe **41** to the supply pipe **45** replenishes the solid target substance **270** delivered by the delivery device **8a** to the molten target container **C3** of the EUV light generation system **11**.

[0118] Accordingly, it is possible to suppress biting of the solid target substance **270** at the edge of the receiving port **801** even when there are variations in the diameter and shape of the solid target substance **270**.

[0119] (2) According to the first embodiment, the stopper **82** moves in a plane intersecting the Z direction, which is the moving direction of the solid target substance **270** passing through the receiving port **801**.

[0120] Accordingly, the direction of the force received by the solid target substance **270** from the stopper **82** is deviated from the Z direction, which is the moving direction of the solid target substance **270** passing through the receiving port **801**. Therefore, it is possible to suppress the solid target substance **270** from being pressed toward the receiving port **801** by the stopper **82**, and suppress biting of the solid target substance **270** from occurring at the edge of the receiving port **801**.

[0121] (3) According to the first embodiment, the stopper **82** moves in the XY plane perpendicular

to the Z direction, which is the moving direction of the solid target substance **270** passing through the receiving port **801**.

[0122] Accordingly, the direction of the force received by the solid target substance **270** from the stopper **82** is significantly deviated from the Z direction, which is the moving direction of the solid target substance **270** passing through the receiving port **801**. Accordingly, it is possible to suppress biting of the solid target substance **270** at the edge of the receiving port **801**.

[0123] (4) According to the first embodiment, the stopper **82** moves in a plane away from the receiving port **801**.

[0124] Accordingly, since there is no receiving port **801** in the surface where the stopper **82** moves, it is possible to suppress the solid target substance **270** from being pressed toward the receiving port **801** by the stopper **82**, and suppress biting of the solid target substance **270** from occurring at the edge of the receiving port **801**.

[0125] (5) According to the first embodiment, the stopper **82** moves by being pressed by the delivery rod **81** when the delivery rod **81** moves forward in the X direction, and moves by the restoring force of the spring **826** when the delivery rod **81** moves backward in the -X direction.

[0126] Accordingly, it is possible to realize the movement of the stopper **82** by the forward and backward movement of the delivery rod **81**.

[0127] (6) According to the first embodiment, the stopper **82** has the first surface **821** that is pressed to the delivery rod **81** via the solid target substance γ when the delivery rod **81** moves forward in the X direction.

[0128] Accordingly, both the solid target substance γ and the stopper **82** can be pressed together with the forward movement of the delivery rod **81**.

[0129] (7) According to the first embodiment, the first surface **821** is inclined with respect to the X direction, which is the direction of the forward movement of the delivery rod **81**.

[0130] Accordingly, it is possible to realize the movement of the stopper **82** in the Y direction by the forward movement of the delivery rod **81** in the X direction.

[0131] (8) According to the first embodiment, the stopper **82** has the second surface **822** that restricts the solid target substance γ pressed by the delivery rod **81** to be away from the first surface **821** from moving in the -X direction, which is the direction of the backward movement of the delivery rod **81**.

[0132] Accordingly, since the solid target substance γ can be suppressed from returning to the -X direction when the delivery rod **81** is returned in the -X direction, one solid target substance **270** can be delivered each time the delivery rod **81** reciprocates.

[0133] (9) According to the first embodiment, the first surface **821** is inclined with respect to the X direction, which is the direction of the backward movement of the delivery rod **81**, and the second surface **822** intersects the X direction at an angle closer to perpendicular thereto than the angle between the X direction and the first surface **821**.

[0134] Accordingly, since the angle between the second surface **822** and the X direction, which is the direction of the forward movement of the delivery rod **81**, is close to a right angle, even when the solid target substance **270** presses the second surface **822**, the stopper **82** can be suppressed from moving toward the outside of the tube **80**.

[0135] (10) According to the first embodiment, the stopper **82** includes the first surface **821** that is pressed via the solid target substance γ when the delivery rod **81** moves forward in the X direction. According to the first embodiment, the delivery rod **81** presses the first surface **821** via the solid target substance γ to move the stopper **82** toward the outside of the tube **80**, and after the solid target substance γ moves away from the first surface **821** of the stopper **82**, the delivery rod **81** is moved backward in the -X direction with the stopper **82** being in contact with the delivery rod **81**.

[0136] Accordingly, the solid target substance **270** may be delivered in the X direction when the delivery rod **81** moves forward in the X direction, and the solid target substance **270** may not be delivered in the -X direction when the delivery rod **81** moves backward in the -X direction.

[0137] (11) According to the first embodiment, by further backward movement of the delivery rod **81** in a state with the stopper **82** stopped after the delivery rod **81** is moved backward in the $-X$ direction to move the stopper **82** to the inside of the tube **80**, a space for receiving the solid target substance β is formed between the receiving port **801** and the facing surface **803** and between the delivery rod **81** and the stopper **82**.

[0138] Accordingly, when the delivery rod **81** is moved backward in the $-X$ direction, the subsequent solid target substance **270** can be received.

[0139] (12) According to the first embodiment, the tube **80** is inclined so as to deliver the solid target substance **270** against gravity when the delivery rod **81** moves forward in the X direction.

[0140] Accordingly, since the solid target substance **270** is delivered against gravity, the height of the entire EUV light generation system **11** including the solid target substance replenishment device **260** can be suppressed, and the degree of freedom in installation can be improved. Since the solid target substance **270** is delivered against gravity not only in the supply pipe **41** but also in the tube **80**, the adjacent solid target substances **270** are conveyed to the discharge port **410** in a state of being in contact with each other, and are pushed out one by one. Therefore, the solid target substance **270** can be stably delivered.

[0141] (13) According to the first embodiment, the stopper **82** moves while pivoting.

[0142] Accordingly, the stopper **82** can be moved by a simple movement mechanism such as the rotation shaft **825**.

[0143] In other respects, the first embodiment is similar to the comparative example.

4. Delivery Device **8b** in which Stopper **82** Moves Linearly

4.1 Configuration and Operation

[0144] FIG. **15** schematically shows the configuration of a delivery device **8b** according to a second embodiment. FIG. **15** is a view of the delivery device **8b** viewed in the Z direction. In the second embodiment, the stopper **82** is supported by the base portion **83** via a slider **828** instead of the rotation shaft **825**, and can move forward and backward linearly in parallel to the Y direction. In the second embodiment as well, the stopper **82** is movable in a plane intersecting the Z direction and movable in a plane away from the receiving port **801**. More preferably, the stopper **82** is movable in a plane perpendicular to the Z direction.

[0145] The rod portion **824** is pressed in the $-Y$ direction by the restoring force of the spring **829**. When the delivery rod **81** moves forward in the X direction, the inclined first surface **821** of the tapered portion **820** is pressed by the delivery rod **81** via the solid target substance γ , and the stopper **82** is moved in the Y direction toward the outside of the tube **80**, thereby forming a passage for moving the solid target substance γ between the receiving port **801** and the facing surface **803** in the tube **80**.

[0146] When the delivery rod **81** is moved backward in the $-X$ direction, the stopper **82** moves toward the inside of the tube **80** in the $-Y$ direction by the restoring force of the spring **829**, but when the stopper **82** abuts the stop pin **827**, the movement of the stopper **82** in the $-Y$ direction is restricted. At this time, the second surface **822** restricts the solid target substance d in the tube **80** from returning in the $-X$ direction. Further, by further movement of the delivery rod **81** in the $-X$ direction, a space for receiving the solid target substance γ supplied from the receiving port **801** is secured between the receiving port **801** and the facing surface **803**.

4.2 Effect

[0147] (14) According to the second embodiment, the stopper **82** moves linearly.

[0148] Accordingly, since the rotation shaft **825** of the stopper **82** is not required, even when fine particles such as scraping debris of the solid target substance **270** are generated, adverse such as effects increasing of the frictional force of the rotation shaft **825** can be suppressed, and the durability of the delivery device **8b** can be improved.

[0149] In other respects, the second embodiment is similar to the first embodiment.

5. Delivery Device **8b** in which First Surface **821** Includes Groove

5.1 Configuration and Operation

[0150] FIG. **16** is a sectional view of a tapered portion **820** of the stopper **82** according to a first modification of the second embodiment. FIG. **16** shows a cross section of a portion of the tapered portion **820** along line I-I of FIG. **15**. The first surface **821** of the tapered portion **820** includes a groove parallel to the XY plane. A reaction force F received by the solid target substance **270** from the first surface **821** when the delivery rod **81** presses the first surface **821** via the solid target substance **270** includes a directional component directing toward an imaginary plane G being parallel to the XY plane and passing through the center of the groove. Therefore, the solid target substance **270** is suppressed from being pressed against the wall surface on the $-Z$ direction side, and biting is suppressed.

[0151] FIG. **17** is a sectional view of the tapered portion **820** of the stopper **82** in a second modification of the second embodiment. In the first modification the groove of the first surface **821** has a V-shaped cross-section. In the second modification, the groove of the first surface **821** may have a cylindrical surface or another concave curved surface, and in other respects the second modification may be similar to the first modification. Such a groove also suppresses biting.

[0152] Here, description is provided on a case that the first surface **821** includes a groove in the second embodiment. However, the first surface **821** may include a groove in the first embodiment.

5.2 Effect

[0153] (15) According to the first and second modifications, the stopper **82** includes the first surface **821** inclined with respect to the X direction, which is the direction of the forward movement of the delivery rod **81**, and the first surface **821** includes the groove in the direction parallel to the XY plane parallel to both the X direction and the Y direction, which is the direction of the movement of the stopper **82**.

[0154] Accordingly, when the delivery rod **81** presses the first surface **821** via the solid target substance **270**, the position of the solid target substance **270** in the direction intersecting the groove can be stabilized.

6. Solid Target Substance Replenishment Device **260c** Including First and Second Detectors **D1**, **D2**

6.1 Configuration

[0155] FIG. **18** shows the configuration of the droplet target generation device **26** of a third embodiment. In the third embodiment, a solid target substance replenishment device **260c** includes first and second detectors **D1**, **D2** and a display unit **57**.

[0156] The first detector **D1** is arranged in the supply pipe **40**. The first detector **D1** may detect a change in the electrical resistance of the supply pipe **40** or may detect a change in vibration. The second detector **D2** is arranged in the vicinity of the discharge port **410**. The second detector **D2** may detect light output from a light source (not shown) and reflected by the solid target substance **270**. The display unit **57** is a device that displays information in a visually recognizable manner, and may be an image display device or a light emitting element.

6.2 Operation

[0157] The first detector **D1** transmits a first detection signal to the target supply processor **55** each time one solid target substance **270** passes through the supply pipe **40**. Each time the first detection signal is received, the target supply processor **55** transmits a drive signal to the drive unit **84** of the delivery device **8a** with a predetermined time difference from the first detection signal. Each time the drive unit **84** receives a drive signal, the drive unit **84** causes the delivery rod **81** to reciprocate once. The drive unit **84** transmits a reciprocal movement signal indicating that the delivery rod **81** has been reciprocated once to the target supply processor **55**.

[0158] The second detector **D2** transmits a second detection signal to the target supply processor **55** each time one solid target substance **270** is discharged from the discharge port **410**. The target supply processor **55** monitors consistency of the reciprocal movement signal and the second detection signal.

[0159] As a first example of the operation of monitoring the consistency, when the reciprocal movement signal is received but the second detection signal is not received, the target supply processor **55** may cause the display unit **57** to display occurrence of an error state. Displaying the error state allows an operator of the EUV light generation system **11** to know that there may be biting or other malfunction in the delivery device **8a** or the supply pipe **41**.

[0160] As a second example of the operation of monitoring the consistency, when the time interval of the second detection signal is shorter than the time interval of the reciprocal movement signal, that is, when the plurality of solid target substances **270** are discharged from the discharge port **410** when the delivery rod **81** reciprocates one time, the target supply processor **55** may cause the display unit **57** to display occurrence of the error state. For example, when the size of the solid target substance **270** varies, the number of solid target substances **270** discharged from the discharge port **410** may be larger than the number of solid target substances **270** delivered from the delivery device **8a**. When the number of solid target substances **270** to be replenished at one time is larger than a desired number, there is a possibility that the capacity of the load lock chamber **C2** is exceeded or a possibility that thermal variation in the molten target container **C3** is increased. Displaying the error state allows the operator of the EUV light generation system **11** to know that there is a possibility that the number of solid target substances **270** to be replenished is too large.

6.3 Effect

[0161] (16) According to the third embodiment, the solid target substance replenishment device **260c** includes the first detector **D1**, the drive unit **84**, and the target supply processor **55**. The first detector **D1** detects the solid target substance **270** passing through the feed device **7** and the supply pipe **40**. The drive unit **84** causes the delivery rod **81** to move forward and backward in parallel to the X direction. The target supply processor **55** controls the drive unit **84** based on the detection result of the first detector **D1**.

[0162] Accordingly, it is possible to match the passing number of the solid target substances **270** passing through the feed device **7** and the supply pipe **40** and the passing number of the solid target substances **270** in the tube **80**.

[0163] (17) According to the third embodiment, the solid target substance replenishment device **260c** includes the second detector **D2** and the target supply processor **55**. The second detector **D2** detects the solid target substances **270** passing through the path from the supply pipe **41** to the supply pipe **45**. The target supply processor **55** determines the consistency between the forward and backward movement parallel to the X direction of the delivery rod **81** and the detection result of the second detector **D2**, and outputs the determination result.

[0164] Accordingly, it is possible to issue a warning when the passing number of the solid target substances **270** in the tube **80** does not match the passing number of the solid target substances **270** in the path from the supply pipe **41** to the supply pipe **45**.

[0165] In other respects, the third embodiment is similar to the first embodiment. Alternatively, in the third embodiment, the stopper **82** may include the delivery device **8b** that moves linearly, or the first surface **821** may include a groove.

7. Others

7.1 EUV Light Utilization Apparatus **6**

[0166] FIG. **19** shows the configuration of the exposure apparatus **6a** connected to the EUV light generation system **11**. In FIG. **19**, the exposure apparatus **6a** as the EUV light utilization apparatus **6** (see FIG. **1**) includes a mask irradiation unit **608** and a workpiece irradiation unit **609**. The mask irradiation unit **608** illuminates, via a reflection optical system, a mask pattern of a mask table **MT** with the EUV light incident from the EUV light generation system **11**. The workpiece irradiation unit **609** images the EUV light reflected by the mask table **MT** onto a workpiece (not shown) arranged on a workpiece table **WT** via a reflection optical system. The workpiece is a photosensitive substrate such as a semiconductor wafer on which photoresist is applied. The exposure apparatus **6a** synchronously translates the mask table **MT** and the workpiece table **WT** to

expose the workpiece to the EUV light reflecting the mask pattern. Through the exposure process as described above, a device pattern is transferred onto the semiconductor wafer, thereby an electronic device can be manufactured.

[0167] FIG. 20 shows the configuration of the inspection apparatus **6b** connected to the EUV light generation system **11**. In FIG. 20, the inspection apparatus **6b** as the EUV light utilization apparatus **6** (see FIG. 1) includes an illumination optical system **603** and a detection optical system **606**. The illumination optical system **603** reflects the EUV light incident from the EUV light generation system **11** to illuminate a mask **605** placed on a mask stage **604**. Here, the mask **605** conceptually includes a mask blanks before a pattern is formed. The detection optical system **606** reflects the EUV light from the illuminated mask **605** and forms an image on a light receiving surface of a detector **607**. The detector **607** having received the EUV light obtains the image of the mask **605**. The detector **607** is, for example, a time delay integration (TDI) camera. A defect of the mask **605** is inspected based on the image of the mask **605** acquired by the above-described process, and for mask suitable manufacturing an electronic device is selected using the inspection result. Then, the electronic device can be manufactured by exposing and transferring the pattern formed on the selected mask onto the photosensitive substrate using the exposure apparatus **6a**.

7.2 Supplement

[0168] The description above is intended to be illustrative and the present disclosure is not limited thereto. Therefore, it would be obvious to those skilled in the art that various modifications to the embodiments of the present disclosure would be possible without departing from the spirit and the scope of the appended claims. Further, it would be also obvious to those skilled in the art that the embodiments of the present disclosure would be appropriately combined.

[0169] The terms used throughout the present specification and the appended claims should be interpreted as non-limiting terms unless clearly described. For example, terms such as “comprise”, “include”, “have”, and “contain” should not be interpreted to be exclusive of other structural elements. Further, indefinite articles “a/an” described in the present specification and the appended claims should be interpreted to mean “at least one” or “one or more.” Further, “at least one of A, B, and C” should be interpreted to mean any of A, B, C, A+B, A+C, B+C, and A+B+C as well as to include combinations of any thereof and any other than A, B, and C.

Claims

1. A solid target substance replenishment device, comprising; a solid target container configured to contain a solid target substance; a first path through which the solid target substance supplied from the solid target container passes; a delivery device including: a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward; and a second path through which the solid target substance delivered by the delivery device is replenished to a molten target container of an extreme ultraviolet light generation apparatus.
2. The solid target substance replenishment device according to claim 1, wherein the stopper moves in a plane intersecting a movement direction of the solid target substance passing through the receiving port.
3. The solid target substance replenishment device according to claim 1, wherein the stopper moves in a plane perpendicular to a movement direction of the solid target substance passing through the

receiving port.

4. The solid target substance replenishment device according to claim 1, wherein the stopper moves in a plane away from the receiving port.

5. The solid target substance replenishment device according to claim 1, wherein the stopper moves by being pressed by the delivery rod when the delivery rod moves forward, and moves by a restoring force of a spring when the delivery rod moves backward.

6. The solid target substance replenishment device according to claim 1, wherein the stopper includes a first surface that is pressed to the delivery rod via the solid target substance when the delivery rod moves forward.

7. The solid target substance replenishment device according to claim 6, wherein the first surface is inclined with respect to a forward movement direction of the delivery rod.

8. The solid target substance replenishment device according to claim 6, wherein the stopper further includes a second surface that restricts the solid target substance pressed by the delivery rod to be away from the first surface from moving in a backward movement direction of the delivery rod.

9. The solid target substance replenishment device according to claim 8, wherein the first surface is inclined with respect to a forward movement direction of the delivery rod, and the second surface intersects the forward movement direction of the delivery rod at an angle closer to be perpendicular thereto than an angle between the forward movement direction of the delivery rod and the first surface.

10. The solid target substance replenishment device according to claim 1, wherein the stopper includes a first surface that is pressed via the solid target substance when the delivery rod moves forward, and the delivery rod presses the first surface via the solid target substance to move the stopper toward the outside of the tube, and after the solid target substance is moved away from the first surface of the stopper, the delivery rod is moved backward with the stopper being in contact with the delivery rod.

11. The solid target substance replenishment device according to claim 1, wherein, by the delivery rod being further moved backward in a state with the stopper stopped after the delivery rod is moved backward into the tube, a space for receiving the solid target substance is formed between the receiving port and the facing surface and between the delivery rod and the stopper.

12. The solid target substance replenishment device according to claim 1, wherein the tube is inclined so as to deliver the solid target substance against gravity when the delivery rod moves forward.

13. The solid target substance replenishment device according to claim 1, wherein the stopper moves while pivoting.

14. The solid target substance replenishment device according to claim 1, wherein the stopper moves linearly.

15. The solid target substance replenishment device according to claim 1, wherein the stopper includes a first surface inclined with respect to a forward movement of the delivery rod, and the first surface includes a groove in a direction parallel to a surface parallel to both a forward movement direction of the delivery rod and a movement direction of the stopper.

16. The solid target substance replenishment device according to claim 1, further comprising: a first detector configured to detect the solid target substance passing through the first path; a drive unit configured to cause the delivery rod to move forward and backward; and a processor configured to control the drive unit based on a detection result of the first detector.

17. The solid target substance replenishment device according to claim 1, further comprising: a second detector configured to detect the solid target substance passing through the second path; and a processor configured to determine consistency between forward and backward movement of the delivery rod and a detection result of the second detector.

18. An extreme ultraviolet light generation apparatus comprising: the solid target substance

replenishment device according to claim 1; the molten target container configured to melt the solid target substance replenished by the solid target substance replenishment device to produce a molten target substance; a nozzle configured to output the molten target substance produced in the molten target container; a laser device configured to irradiate, with pulse laser light, the molten target substance having reached a predetermined region after being output from the nozzle; and an EUV light concentrating mirror configured to concentrate extreme ultraviolet light emitted from plasma generated in the predetermined region.

19. An electronic device manufacturing method, comprising: generating extreme ultraviolet light using an extreme ultraviolet light generation apparatus; the extreme ultraviolet light to an outputting exposure apparatus; and exposing a photosensitive substrate to the extreme ultraviolet light in the exposure apparatus to manufacture an electronic device, the extreme ultraviolet light generation apparatus including: a solid target substance replenishment device including: a solid target container configured to contain a solid target substance; a first path through which the solid target substance supplied from the solid target container passes; a delivery device including: a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward; and a second path through which the solid target substance delivered by the delivery device is replenished to a molten target container of the extreme ultraviolet light generation apparatus; the molten target container configured to melt the solid target substance replenished by the solid target substance replenishment device to produce a molten target substance; a nozzle configured to output the molten target substance produced in the molten target container; a laser device configured to irradiate, with pulse laser light, the molten target substance having reached a predetermined region after being output from the nozzle; and an EUV light concentrating mirror configured to concentrate the extreme ultraviolet light emitted from plasma generated in the predetermined region.

20. An electronic device manufacturing method, comprising: inspecting a defect of a mask by irradiating the mask with extreme ultraviolet light generated by an extreme ultraviolet light generation apparatus; selecting a mask using a result of the inspection; and exposing and transferring a pattern formed on the selected mask onto a photosensitive substrate, the extreme ultraviolet light generation apparatus including: a solid target substance replenishment device including: a solid target container configured to contain a solid target substance; a first path through which the solid target substance supplied from the solid target container passes; a delivery device including: a tube including a receiving port for receiving the solid target substance having passed through the first path and a facing surface facing the receiving port; a delivery rod configured to deliver the solid target substance in the tube by being caused to alternately move forward and backward in a length direction of the tube; and a stopper configured to form a passage through which the solid target substance between the receiving port and the facing surface is moved in the tube by being moved toward an outside of the tube when the delivery rod moves forward, and restrict the target substance in the tube from returning between the receiving port and the facing surface by being moved toward an inside of the tube when the delivery rod moves backward, and a second path through which the solid target substance delivered by the delivery device is replenished to a molten target container of the extreme ultraviolet light generation apparatus; the molten target container configured to melt the solid target substance replenished by the solid target substance replenishment device to produce a molten target substance; a nozzle configured to output

the molten target substance produced in the molten target container; a laser device configured to irradiate, with pulse laser light, the molten target substance having reached a predetermined region after being output from the nozzle; and an EUV light concentrating mirror configured to concentrate the extreme ultraviolet light emitted from plasma generated in the predetermined region.
